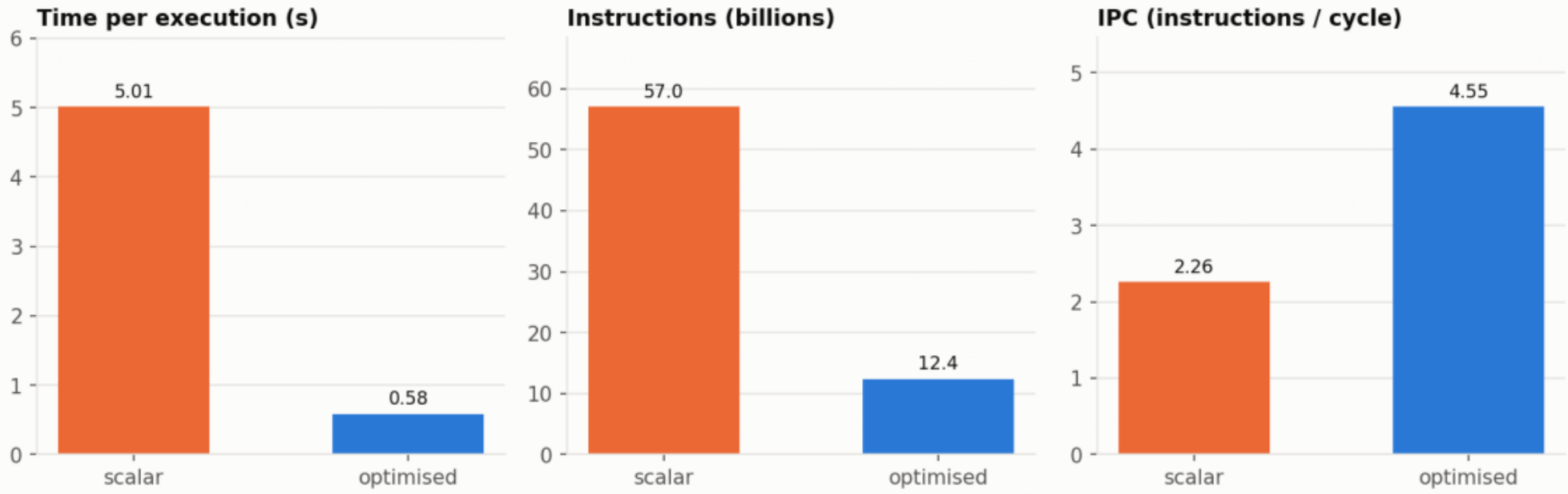
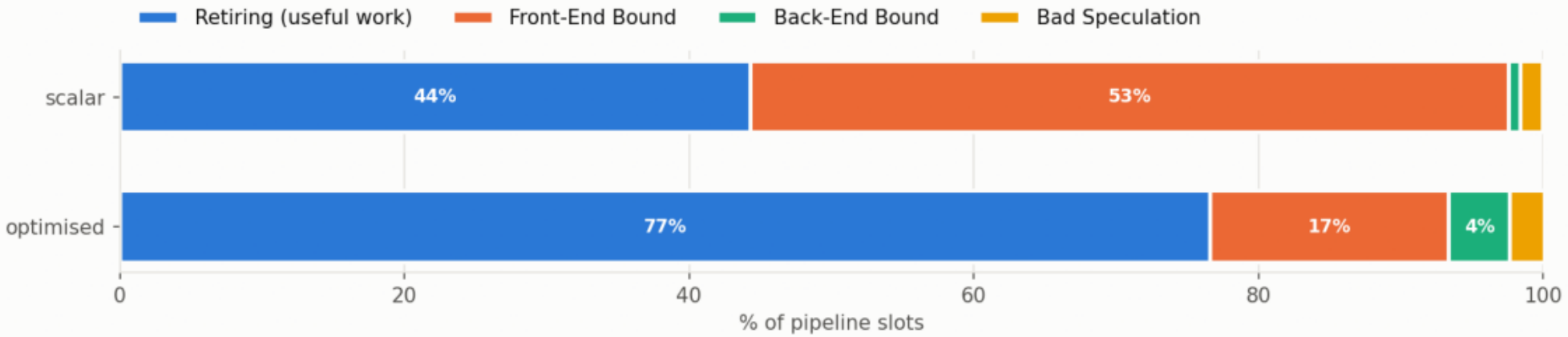
perf - time, instructions and IPC



How to read

Time is the mean of the repeated executions (perf stat -r). Fewer instructions means less work. IPC (instructions per cycle) shows how well the pipeline is fed, bounded by the CPU's issue width; it says nothing about how much work each instruction does, so it does not show vectorisation.

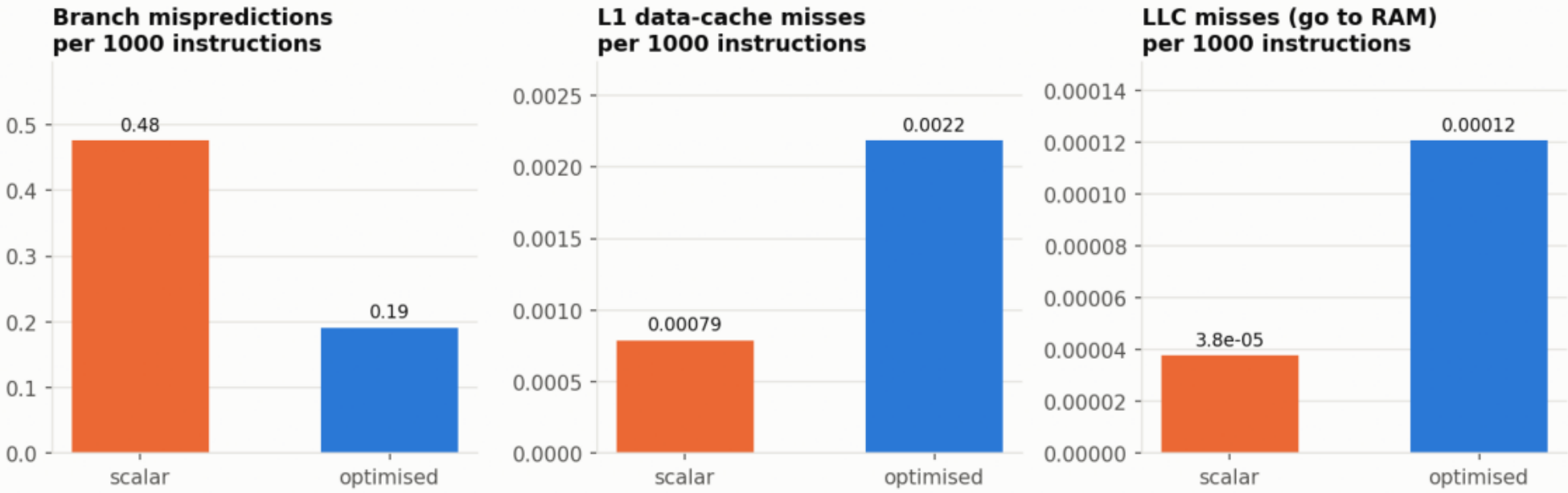
perf - Top-Down bottleneck classification



How to read

Every pipeline slot is sorted into four buckets. Retiring is useful work; the others are losses and say where to look: Front-End Bound (instruction fetch/decode: many small calls, code size), Back-End Bound (waiting for data or execution units), Bad Speculation (mispredicted branches).

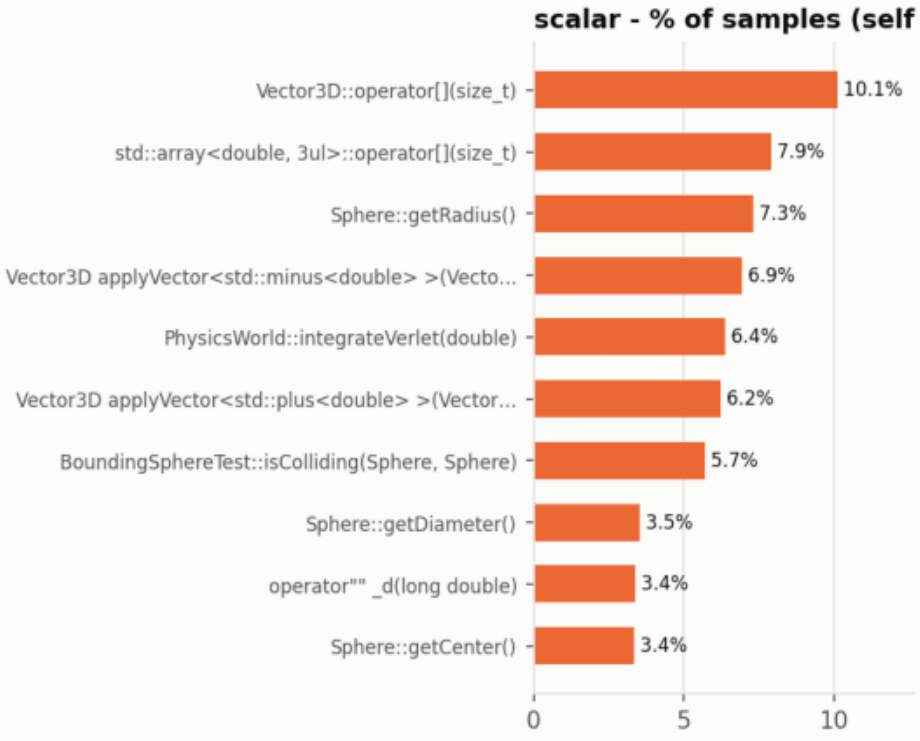
perf - cache and branch misses

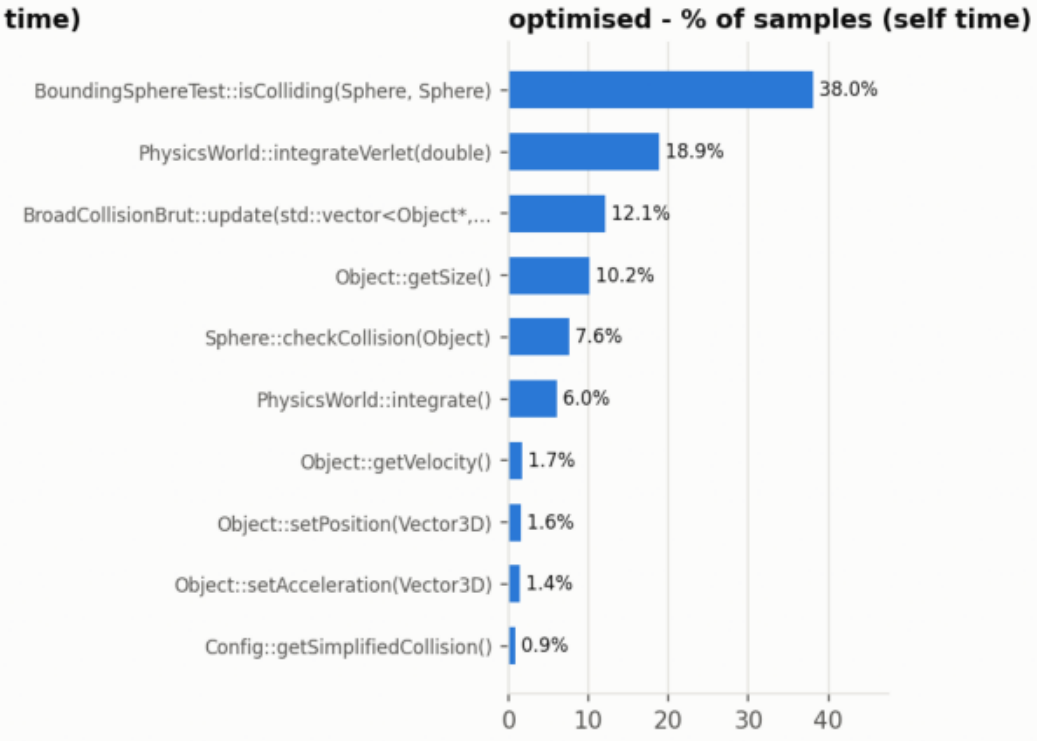


How to read

Misses per 1000 instructions (MPKI), from exact grouped counters. MPKI is preferred over a miss rate: a high rate on a handful of accesses costs nothing. Rough guide: LLC MPKI above 1 (RAM traffic) or branch MPKI above 5 deserve attention.

perf - hottest functions

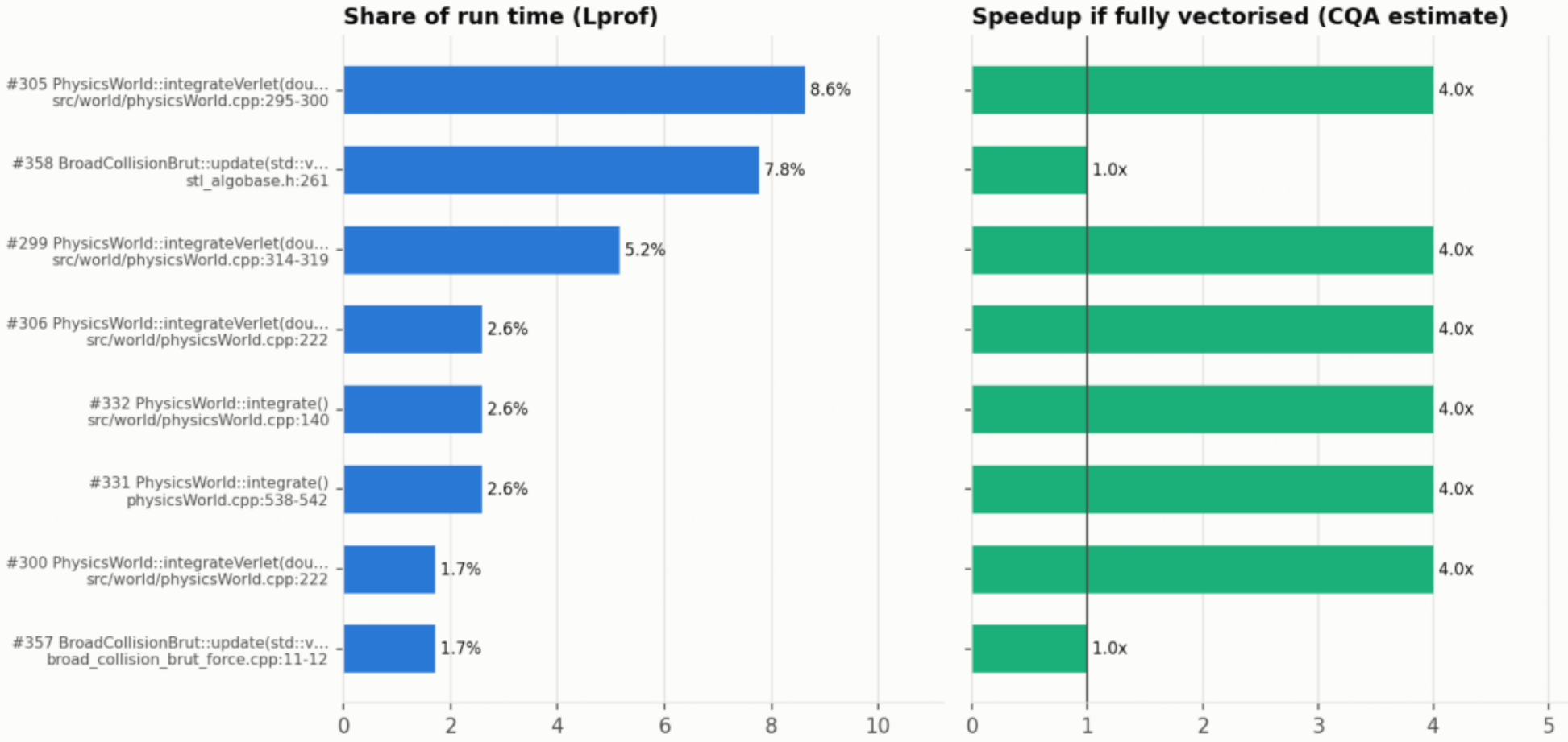




How to read

perf record samples the call stack ~1000 times per second; a function's share of samples is its share of run time (self time, callees excluded). Statistical: small shares are noisy. Trivial getters near the top are calls the compiler could not inline.

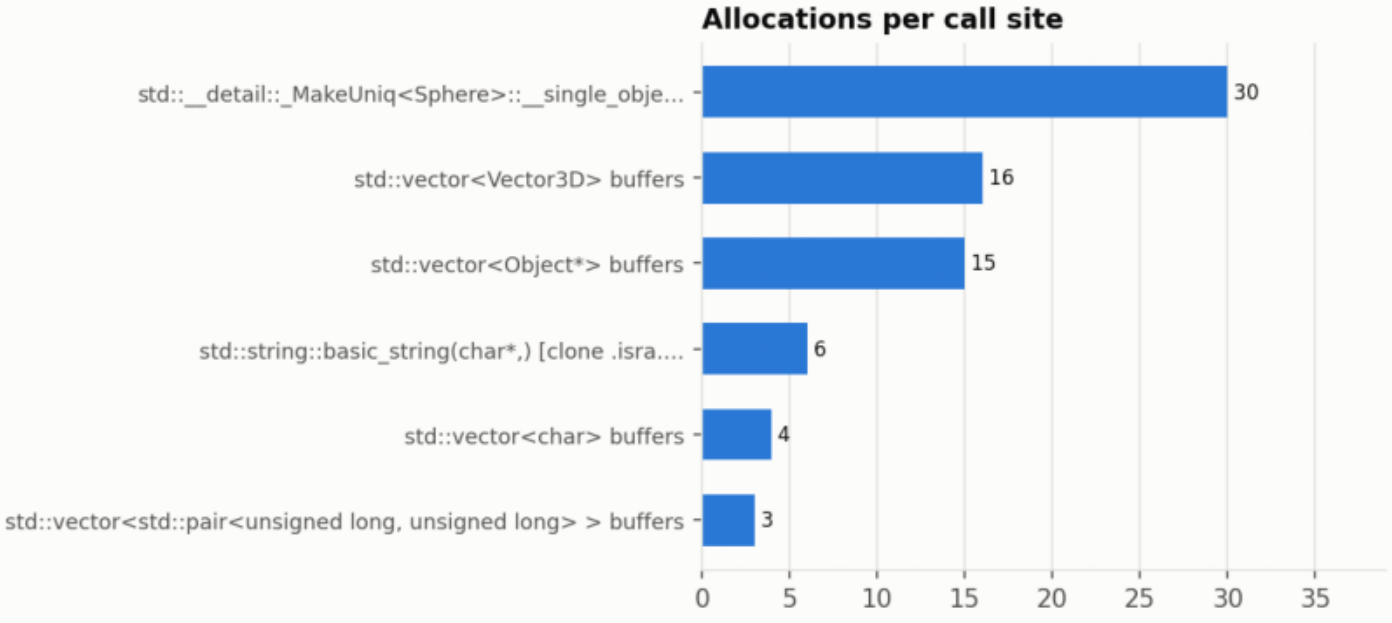
MAQAO - hottest loops and vectorisation potential (optimised build)

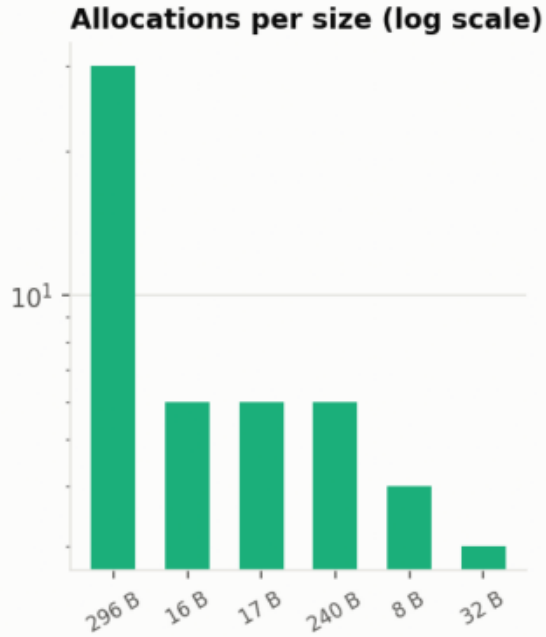


How to read

Left: share of run time per source loop (Lprof, measured). Right: CQA's estimate of the speedup if the same loop used the full vector width (static analysis of the assembly). Loops that are both hot and have a large potential speedup are the first targets.

MALT - where and how often the code allocates

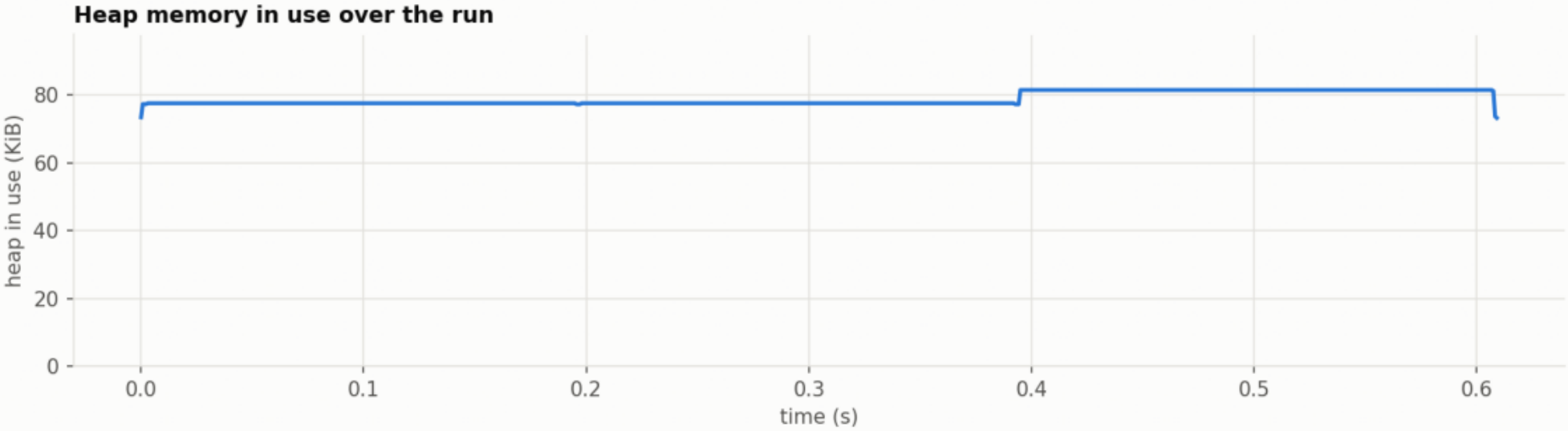




How to read

MALT intercepts every malloc/new, so counts are exact. Left: allocations per call site; right: allocations per size (log scale).
Allocations repeated at every time step are the pattern to remove (reuse the buffer, or a pool allocator).

MALT - heap memory over time



How to read

Bytes requested from the heap and still allocated, sampled over the run. A flat curve means no growth; a rising curve means something accumulates.